TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: CP_SEP490 **Duration time:** From 05/01/2026 To 30/04/2026

Profession: Information Technology **Specialty:** Software Engineering

Kinds of person make registers: Lecturer: ☑ Students: ☐

1. Register information for supervisor

	Full name	Phone	E-Mail	Title
Supervisor 1	Nguyễn Thị Hạnh	0935688515	Hanhnt54@fe.edu.vn	MSc
Supervisor 2				

2. Register information for students

	Full name	Student code	Phone	E-mail	Role in Group
Student 1					Leader
Student 2					Member
Student 3					Member
Student 4					Member
Student 5					Member

3. Register content of Capstone Project

3.1. Capstone Project name:

English**:** Building LocalHub – A Platform for Connecting Local Communities using React, Nodejs, Mongodb.

Vietnamese**:** Xây dựng LocalHub – Nền tảng kết nối cộng đồng địa phương sử dụng React, Nodejs, Mongodb *(SV có thể lựa chọn công nghệ khác).*

Abbreviation**:** PCLC

3.2. Context: (brief introduction)

***Objectives**:*

In the context of rapid digital transformation, the need for **connection among people within the same geographical area** has become increasingly important. However, most existing social media platforms such as Facebook, Zalo, or Instagram focus on **global-scale interaction**, making it difficult for users to share local information, exchange resources, or support each other within their **local neighborhoods or communities**.

LocalHub aims to **build a digital ecosystem for local communities**, fostering human connection, enhancing social interaction, and supporting local socio-economic development at the neighborhood, ward, or township level.

Furthermore, this system can serve as a foundation for **commercial expansion**, including features such as local advertising, nearby business recommendations, or integration of online payments for small-scale local transactions.

Technology/algorithm:

- Front-end: React.js / Next.js/…
- Back-end: ASP.NET Core / Java Spring Boot,…
- Database: MySQL / SQL Server,…
- Version Control: GitHub
- CI/CD Tools:
  - GitHub Actions / GitLab CI/CD (for automated testing and deployment)
  - Docker / Kubernetes (for containerization and scalable deployment)
  - AWS / Firebase / Vercel (for cloud hosting and deployment)

3.3. Summarize the contents to be researched and the expected outputs of the project:

- **Analysis of User Needs and Market Potential**

- Investigate the demand for local communication and information sharing.
- Identify target users such as residents, local businesses, and community managers.
- Analyze existing solutions and their limitations to define LocalHub's competitive advantages.

- **System Requirements and Functional Design**

- Define functional requirements (user registration, posting, chatting, notifications, maps).
- Specify non-functional requirements such as security, scalability, and performance.
- Design user interaction flows and intuitive interfaces for both web and mobile platforms.

- **Technology Research and Selection**

- Evaluate suitable development frameworks (React, Flutter, Node.js, Firebase, etc.).
- Study geolocation, push notification, and real-time data synchronization technologies.
- Consider integration with APIs for maps, authentication, and local advertisements.

- **System Architecture and Database Design**

- Propose an architecture that supports both Web and Mobile clients efficiently.
- Design a database schema that ensures data consistency and fast retrieval.

- Develop admin tools for content moderation and community management.

- **Implementation and Testing**

- Develop and deploy the core modules of the system.
- Test usability, functionality, and performance across multiple devices.
- Ensure system security and protection of user data.

- **Evaluation and Commercialization Potential**

- Assess system performance and user satisfaction.
- Explore business models for monetization, such as local ads or premium business accounts.
- Provide recommendations for future development and scaling.

- Website application: ☑
- Mobile application: ☑ iOT: ☐
- Game: ☐
- Research: ☐ N/A: ☐

3.4. Expected features

1. User Management

User Registration & Authentication:
Allow users to sign up using email, phone number, or social accounts (Google/Facebook).
Profile Management:
Users can create and update personal profiles, including name, avatar, address, and interests.

2. Community Feed and Content Sharing

Post Creation:
Users can create and share posts such as announcements, local events, items for sale, lost and found, or help requests.
Multimedia Support:
Posts may include images, links, and location tags.
Interaction Tools:
Users can like, comment, and share posts within their community.
Categorized Feeds:
Posts can be filtered by category (e.g., “Buy/Sell,” “Jobs,” “Events,” “Emergency,” etc.).

3. Location-Based Features

Geolocation Detection:
Automatically detect the user’s location to provide nearby and relevant content, including posts, events, businesses, and alerts.
Allow posts to include location tags such as points, areas, or full addresses.
Provide an interactive map interface where users can view and explore local posts, events, or businesses, with search options by radius, region, district, or ward.
Enable geofencing to send notifications when a user enters or exits predefined areas, and organize users into local communities based on geographic boundaries (e.g., neighborhoods or ward).

Offer fallbacks for offline usage or low-accuracy location data, such as using city-level positioning when precise coordinates are unavailable.

4. Communication and Interaction

Private Messaging:

Real-time chat between users for inquiries, deals, or discussions.

Notifications System:

Push notifications or in-app alerts for new posts, messages, or updates in the user's area.

Follow and Connect:

Users can follow others to stay updated on their posts or activities.

5. Administration and Moderation

Admin Dashboard:

Interface for administrators to manage users, monitor content, and handle reports.

Content Approval System:

Option to review and approve posts before publishing to maintain community standards.

Violation Reporting:

Users can report inappropriate or fraudulent content.

6. Business and Commercial Features

Local Business Profiles:

Businesses can create pages to promote their shops, services, or products.

Local Advertisements (Local Ads):

Businesses can run paid posts or ads targeted to specific communities.

Premium Accounts:

Subscription-based access for advanced analytics or extended reach.

7. Future Enhancement Possibilities

AI-based recommendation system (suggesting nearby events or posts).

AI-Based Search

AI Community Assistant

AI Content Moderation

Multilingual support for diverse local populations.

Da Nang, 23 /11/2025

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*
Nguyễn Thị Hạnh	